DALES 2 : A Renovated Aerial LiDAR Benchmark for 3D Scene Understanding

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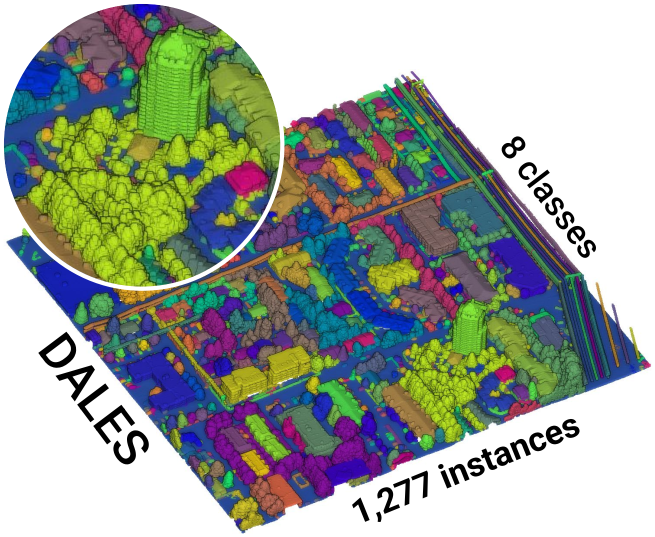
Inria Alteia

Motivations

- Existing 3D benchmarks stop at semantic or instance segmentation.
- No Airborne Laser Scanning (ALS) provides relational ground truth yet.
- **DALES** covers 10km² mixing residential, industrial and vegetated scenes
- Instance annotation artifacts.



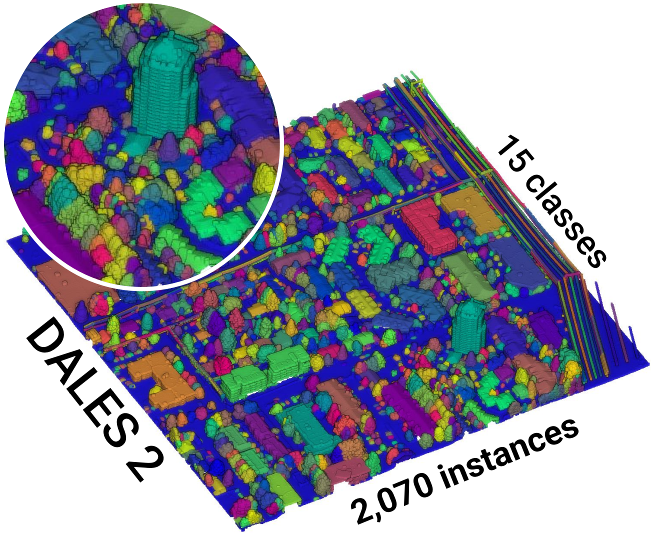
Contributions Overview

DALES 2 : a comprehensive panoptic and hierarchical evolution of DALES.

➔ First ALS-based relational benchmark at scale, designed for utility network analysis and territory management.

High-Fidelity Instances	Extended Taxonomy	Urban Scene Graphs
41,000+ individual trees re-annotated via AI + human pipeline	• 15 classes (up from 8) • Nine new subclasses	• Six edge types • 75,916 edges

*DALES 2 is benchmarked across **semantic segmentation, panoptic segmentation** and **scene graph generation**.*



Tree Instantiation Pipeline

Key insight

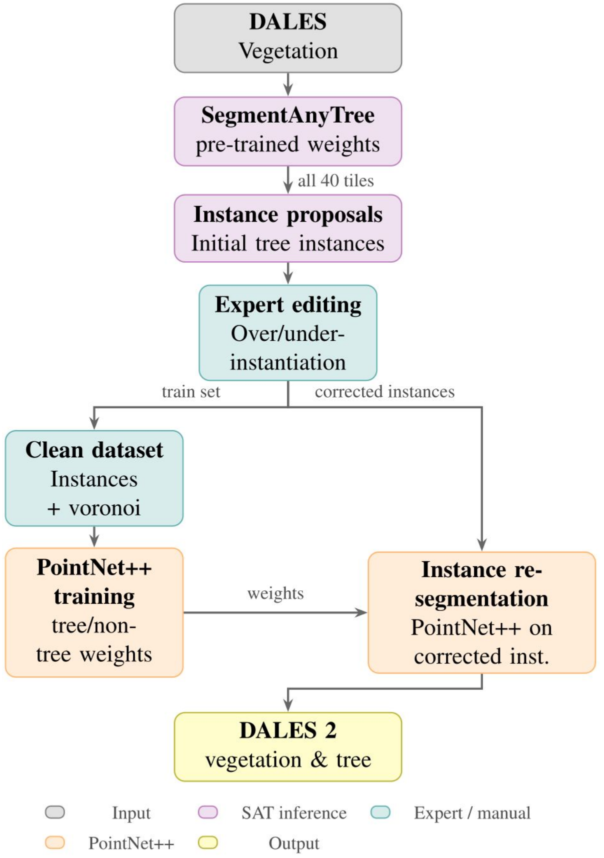
SegmentAnyTree spawns individual *tree* proposals from the *undergrowth* background.

Manual corrections □ label consistency in a 3D editing tool.

Clean *tree* instances and *undergrowth* form *a* training set to further improve delineation through PointNet++.

Key statistics

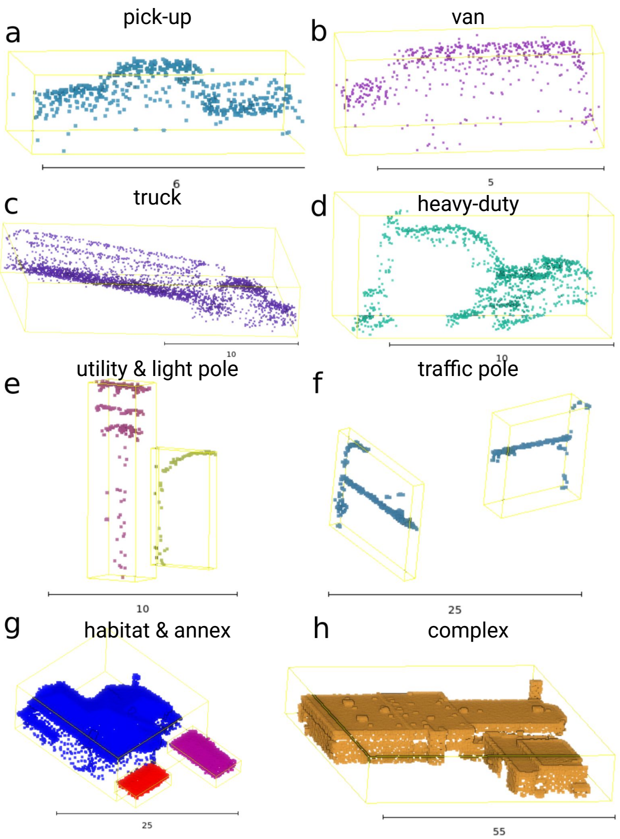
- **41,335** individual trees instantiated
- **125 M** points promoted from “vegetation stuff” to “tree things”
- **55.8%** of legacy instances modified



Hierarchical Taxonomy

Nine novel sub-classes.

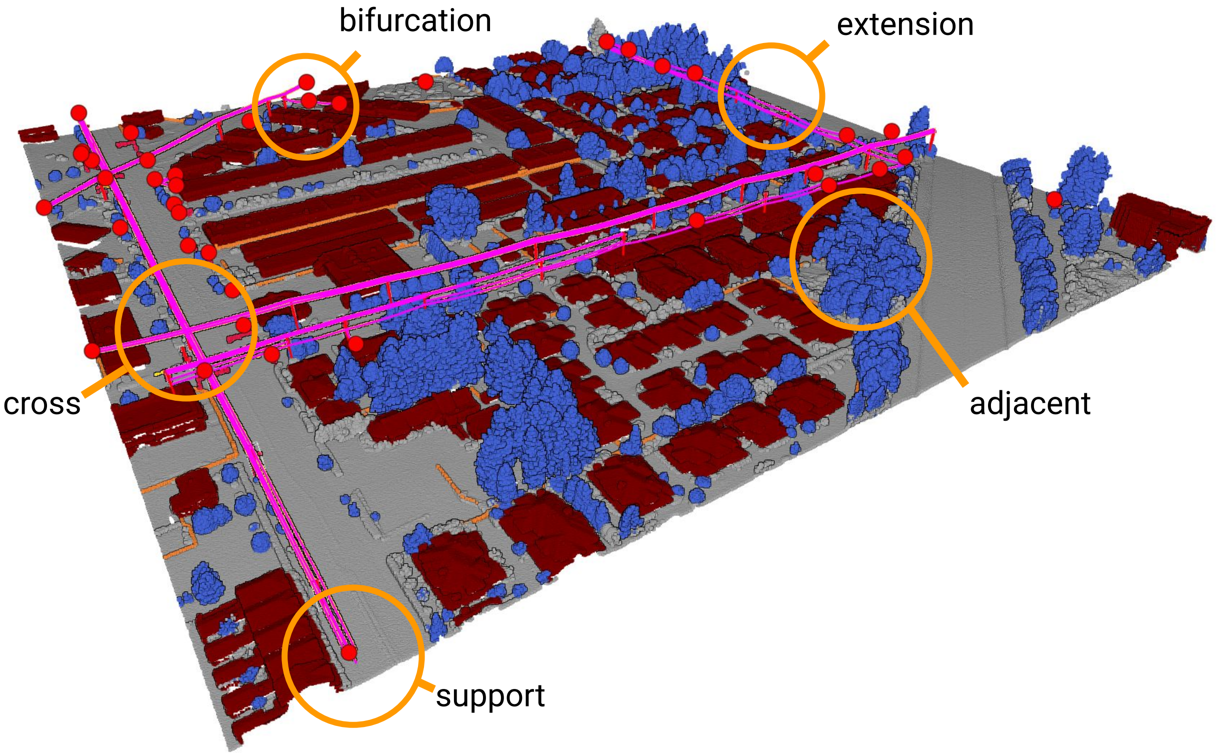
Dataset	Class	#Points	#Instances	Avg pts/inst	Median pts/inst
DALES 2	Ground†	→ 247 M	-	-	-
	Vegetation†	↓ 34 M	-	-	-
	Car	↓ 2.7 M	10,411	259	262
	Powerline	→ 996 K	4,145	240	141
	Fence†	→ 2.1 M	-	-	-
	Tree *	↑ 125 M	41,335	3 K	1.2 K
	Pick-up *	↑ 453 K	1,132	400	398
	Van & Truck *	↑ 906 K	1,081	838	633
	Heavy-duty *	↑ 118 K	145	815	682
	Utility Pole *	↑ 135 K	536	252	182
	Light Pole *	↑ 81 K	473	172	153
	Traffic Pole *	↑ 47 K	166	281	250
	Habitat *	↑ 53 M	5,319	10 K	9.4 K
	Complex *	↑ 23 M	392	60 K	38 K
	Annex *	↑ 2 M	1,098	1.9 K	1.7 K



Scene Graphs

75,916 total edges

~15K infrastructure
~61K proximity



Benchmarking Results

Semantic segmentation

Model	Annex	Powerline	Heavy-duty	...	mIoU
RandLA-Net	**8.9**	**52.3**	**1.0**	...	**29.2**
KPConv	18.3	60.9	**0.1**	...	42.3
PTv3	**56.9**	**96.7**	**34.4**	...	**68.9**

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Panoptic quality

Model	Annex	Powerline	Heavy-duty	...	Mean PQ
OneFormer3D	46.0	38.4	9.2	...	46.9
SuperCluster	56.1	2.7	25.2	...	45.3

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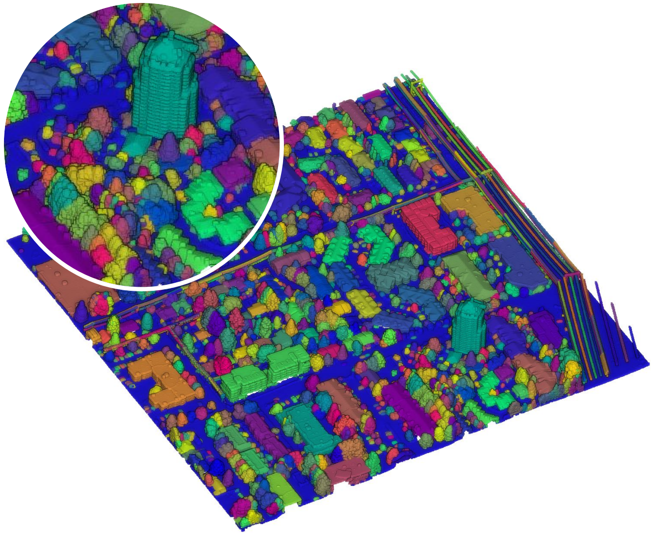
Edge retrieval

Relation	F1
Extension	96.4
Bifurcation	16.1
Cross	68.1
Support	99.7
Adjacent	69.6
Near	59.8
No edge	54.1

Conclusion & Takeaways

- **First ALS benchmark with panoptic + relational ground truth jointly**
- **41K trees** re-instantiated via AI-assisted pipeline
- **75K scene graph edges** across **6 typed relations** at urban scale
- Graph editing tool released for community extension



Open Challenges

- Bifurcation detection at network branch points
- Proximity relation ambiguity in dense urban point clouds
- Multi-scale learning for small vertical structures (poles, annexes)